

Internship Report

Project Title

Twitter Analytics Dashboard using Power BI

Submitted By

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Internship Domain

Data Analytics

Organization

ElevanceSkills

Submission Date

02-06-2026

Tools Used

Power BI, Microsoft Excel

Introduction

The internship project focused on the development of an interactive Twitter Analytics Dashboard using Power BI. The main objective of the project was to analyze Twitter engagement data and convert raw datasets into meaningful visual insights through business intelligence techniques. The dashboard was designed to help understand user engagement patterns, tweet performance, media interactions, and overall audience activity across different time periods.

In today's digital environment, social media platforms generate large amounts of data every second. Organizations and analysts use this data to understand audience behavior, measure engagement, improve marketing strategies, and make informed business decisions. Twitter analytics is particularly important because it provides insights into how users interact with tweets through likes, retweets, replies, impressions, and media engagement. This internship project provided practical exposure to analyzing such social media metrics using Power BI.

The project involved importing datasets, cleaning and transforming the data, creating calculated columns and measures using DAX, designing multiple analytical visualizations, and integrating them into a professional dashboard layout. Several conditions and advanced filters were implemented as part of the internship tasks, including time-based visibility logic, odd-even filtering conditions, word count restrictions, and weekday filtering.

One of the primary goals of the internship was to understand how dashboards are designed in real business environments. The dashboard was therefore developed not only for analytical accuracy but also for visual presentation and user interaction. KPI cards, slicers, tooltips, trend charts, and conditional formatting were used to improve dashboard usability and readability.

The internship also provided practical experience in project documentation and hosting. The completed dashboard project, screenshots, dataset, and supporting documentation were uploaded to GitHub and organized professionally for submission purposes. Overall, this project helped strengthen technical knowledge in Power BI and improved understanding of business intelligence workflows and dashboard reporting practices.

Background

The internship project was based on a Twitter dataset containing several engagement-related metrics associated with tweets posted on the platform. The dataset included information such as likes, retweets, replies, tweet impressions, media engagements, media views, engagement rates, tweet dates, app opens, and user profile interactions. The project required converting this raw dataset into an interactive analytical dashboard capable of providing meaningful insights regarding tweet performance and audience behavior.

Social media analytics has become an important component of digital marketing and business intelligence. Organizations use platforms such as Twitter to promote products, communicate with customers, and track public engagement. However, raw data alone is difficult to interpret without proper visualization and analysis. Business intelligence tools such as Power BI help transform raw data into interactive dashboards that support data-driven decision-making. This internship project was therefore designed to provide practical exposure to the process of data visualization and analytical reporting.

The dashboard development process involved multiple stages. Initially, the dataset was imported into Power BI and cleaned using transformation techniques. Several calculated columns and DAX measures were then created to support analytical requirements such as conditional filtering and KPI calculations. Different chart types including line charts, clustered bar charts, and dual-axis charts were used to represent engagement trends and comparisons effectively.

The internship also emphasized dashboard design principles. Attention was given to color consistency, chart readability, layout organization, typography, spacing, and dark-theme dashboard styling. Interactive elements such as slicers and tooltips were implemented to improve the user experience and allow dynamic filtering across visuals.

The completed dashboard provides a clear overview of Twitter engagement behavior and demonstrates how business intelligence tools can be used to extract meaningful information from social media datasets. The project reflects practical understanding of data analysis, visualization, dashboard development, and reporting techniques commonly used in the analytics industry.

Learning Objectives

The internship project was designed to improve both technical and analytical skills related to business intelligence and dashboard development. One of the primary learning objectives was to understand how Power BI can be used to transform raw datasets into interactive visual reports capable of supporting business analysis and decision-making.

A major objective of the project was to gain practical experience in Power BI dashboard development. Before starting the internship, the understanding of dashboard design and data visualization was limited to basic concepts. Through this project, knowledge was gained regarding importing datasets, creating relationships, cleaning data, designing visualizations, and applying formatting techniques to improve dashboard presentation.

Another important learning objective was to understand DAX calculations and conditional logic implementation. Several internship tasks required the use of calculated columns and measures to apply conditions such as even-number filtering, odd-date filtering, time-based visual visibility, and word count restrictions. These tasks helped improve understanding of DAX formulas and analytical logic creation within Power BI.

The internship also focused on improving data storytelling skills. The objective was not only to display numbers but also to present insights in a visually meaningful and understandable way. Different chart types were selected based on analytical requirements, and dashboard components such as KPI cards, slicers, tooltips, and insight sections were used to improve communication of information.

Finally, the project aimed to improve project documentation and hosting skills. The completed dashboard was documented using a professional README file, screenshots were organized systematically, and the project files were uploaded to GitHub and Google Drive. This helped improve understanding of project presentation and portfolio preparation practices commonly expected in analytics and technical domains.

Overall, the internship successfully achieved its learning objectives by providing practical exposure to Power BI, dashboard development, data visualization, analytical thinking, and professional project presentation.

Activities and Tasks

The internship project involved multiple activities related to data analytics, dashboard design, and business intelligence reporting using Power BI. The project workflow began with importing the Twitter dataset into Power BI and understanding the structure of the data. The dataset contained multiple engagement-related fields including likes, retweets, replies, impressions, media engagements, media views, tweet text, tweet dates, engagement rates, and profile interaction metrics. Before creating visualizations, it was necessary to inspect the dataset carefully and identify relevant columns required for analysis.

Dashboard styling and layout organization were also important activities during the internship. The initial dashboard designs appeared cluttered and inconsistent, so considerable effort was spent improving the visual appearance of the dashboard. A dark-themed dashboard style was selected to create a modern and professional appearance. Chart sizes, colors, typography, borders, and spacing were adjusted carefully to maintain consistency across all visuals. Interactive components such as KPI cards, slicers, tooltips, and trend indicators were integrated to improve user interaction and readability.

Another important activity involved creating KPI cards for displaying key metrics such as total engagement, total likes, total retweets, total impressions, and average engagement rate. These KPI cards helped summarize the overall performance of the Twitter dataset and improved the executive dashboard experience. Slicers were also implemented to provide interactive filtering functionality across the dashboard.

The internship project additionally involved organizing and documenting the completed work professionally. Screenshots of dashboard visuals were captured and uploaded to GitHub along with the Power BI project file, dataset, and README documentation. The project repository was organized systematically using folders for screenshots, datasets, and project files. This activity helped improve project presentation and documentation skills.

Overall, the activities performed during the internship provided valuable practical experience in Power BI dashboard development, data transformation, business intelligence reporting, and analytical problem-solving.

Skills and Competencies

The internship project helped develop both technical and professional competencies related to data analytics and business intelligence. One of the most important technical skills improved during the internship was Power BI dashboard development. Through continuous work on visualizations, filters, slicers, KPI cards, and layout organization, practical understanding of dashboard creation improved significantly. The project provided hands-on experience with creating interactive reports capable of presenting complex data in a visually understandable manner.

Another major technical competency developed during the internship was DAX calculation and measure creation. Several internship tasks required the implementation of calculated columns and conditional logic using DAX formulas. Examples included filtering even tweet impressions, odd tweet dates, word count conditions, and time-based visual visibility logic. Writing and testing these calculations helped strengthen understanding of analytical expressions and logical conditions in Power BI.

Communication and documentation skills were also strengthened through the project. The internship required organizing project files, preparing GitHub documentation, capturing dashboard screenshots, and writing a professional internship report. These activities helped improve understanding of project presentation practices commonly expected in professional environments.

Another competency developed during the internship was understanding interactive reporting techniques. The use of slicers, tooltips, KPI cards, and conditional visual interactions demonstrated how dashboards can be designed to provide dynamic and user-friendly analytical experiences. This improved understanding of how business intelligence tools support decision-making processes through interactive data exploration.

Overall, the internship project contributed significantly to the development of technical knowledge, analytical thinking, visualization skills, and professional project management abilities. The practical experience gained through the internship will be highly valuable for future data analytics and business intelligence projects.

Feedback and Evidence

The internship project successfully resulted in the development of a fully interactive Twitter Analytics Dashboard using Power BI. The completed dashboard integrated all internship tasks into a professional analytical reporting environment and demonstrated practical application of business intelligence concepts. The project served as evidence of understanding in areas such as data visualization, dashboard interaction, DAX calculations, analytical reporting, and project documentation.

One of the most important outcomes of the project was the successful implementation of six different analytical tasks within the same dataset and reporting environment. Each task focused on a different analytical objective and required specific filters, calculations, and visualization techniques. The final dashboard included interaction analysis, engagement comparisons, trend monitoring, media engagement analysis, and top-performing tweet identification. These visualizations collectively demonstrated the ability to interpret social media data and present insights effectively.

Another important piece of evidence was the GitHub repository created for the project. The repository contained the Power BI dashboard file, dataset, screenshots of visualizations, and project documentation. Organizing the project files systematically and uploading them to GitHub demonstrated professional documentation and project presentation practices. The repository served as a portfolio-style representation of the internship work completed during the project.

Throughout the dashboard development process, several improvements were made based on self-evaluation and iterative design adjustments. Visual layouts were reorganized, chart readability was improved, tweet labels were shortened for clarity, and dashboard themes were standardized to achieve a professional appearance. These iterative improvements reflected growing understanding of dashboard usability and design principles.

The final dashboard successfully met the internship requirements by combining analytical functionality, interactivity, and professional presentation. The project demonstrated practical understanding of business intelligence workflows and provided measurable evidence of technical and analytical skill development during the internship period.

Challenges and Solutions

During the internship project, several technical and design-related challenges were encountered while developing the Twitter Analytics Dashboard. These challenges provided valuable learning opportunities and helped improve problem-solving abilities, analytical thinking, and technical understanding of Power BI dashboard development. Addressing these issues required experimentation, logical reasoning, and continuous refinement of the dashboard design and functionality.

One of the first challenges encountered during the project was managing data cleaning and formatting issues within the dataset. Some date fields and text columns required proper formatting before they could be used effectively in visualizations and calculations. Inconsistent formatting initially caused errors during chart creation and filtering processes. This challenge was addressed by using Power Query transformations and DAX calculations to clean the data, standardize formats, and create additional calculated columns for analytical purposes.

Another significant challenge involved implementing advanced filtering logic required by the internship tasks. Several tasks required unusual conditions such as selecting only even tweet impressions, odd tweet dates, excluding weekend tweets, restricting word counts, and controlling visual visibility during specific time intervals. Applying multiple filters simultaneously often resulted in missing data situations or empty visualizations. To solve this issue, calculated columns and DAX measures were created carefully to handle each condition separately.

Handling missing or empty data conditions also became a major challenge during dashboard development. Some filtering combinations produced no qualifying records, resulting in blank charts that appeared visually broken. Instead of removing the conditions, professional dashboard practices were applied by displaying explanatory messages such as “No matching data for current filters.”

Dashboard layout organization was another difficult aspect of the project. Initially, combining all six analytical tasks into a single dashboard page resulted in overcrowding and inconsistent spacing between visuals. Different charts had varying dimensions, text sizes, and alignment issues that reduced the overall professional appearance of the dashboard. This challenge was solved by redesigning the layout structure completely. KPI cards were aligned consistently, slicers were repositioned, chart sizes were standardized, and visual spacing was improved to create a cleaner and more balanced dashboard design.

Overall, the challenges encountered during the internship project played a major role in improving technical understanding and analytical problem-solving skills. Successfully overcoming these difficulties provided practical exposure to real-world dashboard development workflows and strengthened confidence in handling business intelligence projects independently.

Outcomes and Impact

The internship project successfully achieved its primary objective of developing an interactive Twitter Analytics Dashboard using Power BI. The completed dashboard transformed raw Twitter engagement data into meaningful visual insights and demonstrated practical understanding of data analytics, dashboard development, and business intelligence concepts. The project produced several positive outcomes that contributed to both technical growth and analytical skill development.

One of the most significant outcomes of the internship was the improvement in practical knowledge of Power BI. Before the project, understanding of dashboard development was limited to basic concepts and simple visualizations. Through the implementation of multiple internship tasks, experience was gained in importing datasets, cleaning data, creating DAX calculations, designing visualizations, applying conditional filters, and organizing interactive dashboards.

The internship project also contributed significantly to improving dashboard design and visualization skills. Through repeated styling improvements, better understanding was developed regarding layout organization, color consistency, spacing, typography, and chart readability. The final dashboard design achieved a more professional appearance with a modern dark-themed interface and structured visual hierarchy.

Another important impact of the project was the improvement in project documentation and portfolio preparation skills. The dashboard project was documented professionally using a README file, organized GitHub repository, screenshots, and supporting dataset files. This process improved understanding of how technical projects should be presented and hosted for professional or academic submissions.

The internship additionally improved understanding of interactive reporting concepts. The implementation of slicers, tooltips, KPI cards, and conditional visibility logic demonstrated how dashboards can be designed to provide dynamic analytical experiences for users. This exposure helped develop a stronger understanding of user-focused dashboard development practices.

From a professional perspective, the project increased confidence in working independently on analytical assignments and business intelligence tasks. Completing multiple internship tasks within the same dashboard environment demonstrated the ability to handle structured project requirements and deliver analytical outputs within specified conditions.

Overall, the internship project created a strong foundation in Power BI dashboard development and data analytics. The knowledge and practical experience gained through this project will be highly valuable for future academic projects, internships, and professional opportunities in the fields of data analytics and business intelligence.

Conclusion

The internship project on Twitter Analytics Dashboard development provided valuable practical experience in data analytics, business intelligence, and interactive reporting using Power BI. Throughout the internship, multiple analytical tasks were completed successfully, including engagement analysis, trend monitoring, interaction comparison, media engagement tracking, and top-performing tweet identification. The project helped transform theoretical understanding of analytics concepts into practical implementation through real dashboard development activities.

One of the most important achievements of the internship was the successful creation of a professional and interactive Power BI dashboard. The dashboard integrated multiple analytical visualizations, KPI cards, slicers, filters, and conditional logic into a single reporting environment. Considerable attention was given to dashboard organization, readability, and user interaction to ensure that the final output was both informative and visually professional.

The internship also helped improve technical knowledge in several areas, including DAX calculations, data transformation, dashboard design, and business intelligence workflows. Through continuous experimentation and problem-solving, understanding of Power BI functionality and analytical reporting techniques improved significantly. Challenges encountered during the project, such as complex filtering conditions, layout management, and data handling issues, contributed positively to learning and skill development.

Another valuable aspect of the internship was the development of professional project documentation and presentation practices. The completed dashboard, screenshots, dataset, and README documentation were organized systematically using GitHub and Google Drive. This experience improved understanding of how technical projects should be prepared and presented for professional evaluation and portfolio purposes.

The project additionally strengthened analytical thinking and decision-making skills. By analyzing engagement data and creating meaningful visual insights, greater understanding was developed regarding how dashboards support business analysis and data-driven decision-making processes. The use of KPI cards, slicers, and interactive visualizations demonstrated the importance of presenting data in a way that improves usability and interpretation.

Overall, the internship project was highly beneficial in improving both technical and professional competencies related to data analytics and business intelligence. The experience gained during this internship has created a strong foundation for future projects involving Power BI, dashboard development, and analytical reporting. The project successfully achieved its objectives and provided meaningful practical exposure to real-world data visualization and business intelligence practices.